Shot Transition Detection Using a Perceptual Distance Function

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Abstract

A shot transition occurs when our eyes fail to maintain perceptual continuity. To maintain perceptual continuity, the majority of perceptual features should be similar from one frame to its next. In this paper, we propose using the dynamic partial function (DPF) as the inter-frame distance function to detect perceptual discontinuity. Through analysis and experiments, we show that DPF is more effective than traditional distance functions for detecting shot transitions.

1 Introduction

Video *shot-transition* detection is a critical research topic, since reliably recovering a *shot*, the elementary video unit, provides the foundation for nearly all high-level video segmentation/analysis algorithms [2]. There has been a large body of work on this research topic. A typical shot-transition detecting technique involves in two stages:

1. *Extracting intra-frame features.* Intra-frame features are features of individual video frames (i.e., still images). As depicted in [3], widely used image features include colors, intensities, textures, shapes, and layout information.
2. *Comparing inter-frame distance.* Videos differ from still images in that they carry both spatial changes in time and temporal changes in space. A shot transition takes place at where inter-frame distance is *perceptually significant.*

Given intra-frame features, the heart of the shot-transition detection research is to devise a perceptual distance function which can reliably capture the perceptual discontinuity at shot boundaries. We use Figure **??** to illustrate what a good distance function

should accomplish versus what a poor distance function might result. The x-axis of the figure depicts the frame number; the y-axis the distance between the ith and the $i + 1^{th}$ frames. Figure **??**(a) shows a poor distance function, which fails to separate the perceptually dissimilar frames (shot boundaries) from the perceptually similar ones. Figure **??**(b) shows a good distance function, which successfully separates the large distances at the shot boundaries from the small distances at the other frame boundaries.

In this paper, we propose using the dynamic partial function [1] (DPF) for detecting shot boundaries. As its name suggests, DPF *dynamically* selects a *partial* feature subset to compare frames. DPF works effectively for the following two reasons:

1. *Partial.* Two frames that belong to the same shot should have significant number of similar intra-frame features for maintaining our perceptual continuity. Object movement and camera motions affect only a subset of the features. Once these features are discounted by DPF, the remaining features should be perceptually similar, or a shot transition takes place between the frames.

2. *Dynamic.* DPF always discounts the largest feature distances, and thus it can dynamically ignore the influence of different features caused by different spatial or temporal changes. For instance, a *pan* might alter just color features. A *zoom* might affect just texture channels. DPF dynamically removes largest feature distances, and hence it removes those features that do not help maintain perceptual continuity.

In summary, our perception maintains continuity when there exist sufficient number of similar features from one frame to its next. DPF removes the largest dissimilar features. If the remaining features are not similar, our perception continuity cannot be maintained, and therefore a shot transition takes place. Our empirical study shows that DPF achieves superior shot-detection accuracy over traditional distance functions.

The rest of this paper is organized into four sections. Section 2 presents DPF. Section 3 summarizes the experimental results. Finally, we offer out concluding remarks in Section 4.

2 Dynamic Partial Function

Given a set of intra-frame features, our goal is to design a distance function that can accurately determine shot boundaries.

The Minkowski metric is widely used in shot boundary detection algorithms. Suppose two objects X and Y are represented by two p dimensional vectors $(x_1, x_2, \cdots, x_p)$ and $(y_1, y_2, \cdots, y_p)$, respectively. The Minkowski metric $d(X, Y)$ is defined as

$$d(X, Y) = (\sum_{i=1}^{p} |x_i - y_i|^r)^{\frac{1}{r}}, \tag{1}$$

where r is the Minkowski factor for the norm. Particularly, when r is set as 2, it is the well known Euclidean distance; when r is 1, it is the Manhattan distance (or L_1 distance). An object located a smaller distance from a query object is deemed more similar to the query object.

Measuring similarity by the Minkowski metric is based on one assumption: two frames are similar when the aggregated distance between all their features is close. However, this assumption runs counter to how our eyes maintains perceptual continuity. The perceptual continuity is maintained because *constancy* is maintained in a large fraction of the feature-channels. For instance, feature vector $V_1 = (0.5, 0.5, 0.1)$ and feature vector $V_2 = (0.5, 0.5, 0.9)$ can have good perceptual continuity since the majority of the features are stable. On the other hand, feature vector $V_3 = (0.3, 0.3, 0.2)$ and feature vector $V_4 = (0.5, 0.5, 0.9)$ is perceptually discontinuous, since all features are different. (Frame V_4 has no stable reference on frame V_3.) However, the Euclidean distance would characterize that the V_3 and V_4 pair is more similar to the V_1 and V_2 pair.

Based on the observations explained above, we designed a distance function to better characterize perceptual continuity. We define the *feature distance* on the ith feature as

$$\delta_i = |x_i - y_i|. \qquad (i = 1, \cdots, p)$$

We define sets Δ_m as

$$\Delta_m = \{The \;\; smallest \;\; m \; \delta's \;\; of \;\; (\delta_1, ..., \delta_p)\}.$$

DPF can then be expressed as

$$d(m, r) = (\sum_{\delta_i \in \Delta_m} {\delta_i}^r)^{\frac{1}{r}}. \tag{2}$$

DPF has two adjustable parameters: m and r. Parameter m can range from 1 to p. When $m = p$, it degenerates to the Minkowski metric. When $m < p$, it counts only the smallest m feature distances between two frames, and the influence of the $(p - m)$ largest feature distances is eliminated. Note that DPF dynamically selects features to

be considered for different pairs of frames. This is achieved by the introduction of Δ_m, which changes dynamically for different pairs of frames. In Section 3, we will show that if a proper value of m is chosen, we can better single out frames that cause perceptual discontinuity.

3 Experiments

Our testbed consists of 150 video clips downloaded from the Internet. The videos cover the following subjects:

- Cartoon. 30 clips, each clip lasts for 50 seconds.
- Documentary: 70 clips, each lasts for two to five minutes [].
- Comedy. 50 clips, each lasts for up to 30 seconds [].

For echaracterizing a frame, we extract a set of 144 intra-frame features. These features include color histograms, color means, color variances, color spreadness, color-blob elongation, and texture features in three orientations (vertical, horizontal, and diagonal) and three resolutions (coarse, medium, and fine) [].

Our experiments have two goals. The first one is to find the optimal parameter m settings for DPF (Section 3.1). The second goal is to compare the shot detection accuracy between employing DPF and employing the Minkowski metric as the inter-frame distance function (Section 3.2).

3.1 Parameter m

To find a good m, we use a machine learning approach. We sample 40% of the video clips as the training data to discover a good m. We then use the remaining 60% video clips as testing data to examine the effectiveness of the learned m.

For the training data, we label the accurate positions of shot boundaries. We than experiment

1. Report the optimal range for all three video types.
2. Select an interset range.

YI: Here, we want to plot two graphs for $m = 144$ and $m =$ optimal to show that the inter-frame distances at the shot boundaries are much better separated by the optimal m. We plot three sets of charts, each for one type of videos. Note that you will generate 6 figures. Do not put them into one figure.

3.2 DPF vs. Minkowski

Report the same results as in the old text.

4 Conclusion

Most of the previous shot detection algorithms use Minkowski-like metric as the inter-frame distance function. In this paper, we explained that the Minkowski-like metric may not be able to capture perceptual discontinuity most effectively for shot-transition detection. We introduced DPF, which discounts the changes (which likely to be spatial and temporal changes) between frames. If the remaining features are still quite different, then perceptual continuity may not likely exist between the two frames, and hence it is declared a shot boundary. We use two different sets of intra-frame features and show that DPF outperforms the Minkowski metric in precision/recall on different types of video clips.

References

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